

Technical Documentation

Project Overview

Project Title: Multi-Site Enterprise Network Implementation (Project E).
Objective: To implement a secure, scalable network infrastructure utilizing Variable Length Subnet Masking (VLSM), multi-tier routing, and centralized application services.

Section 1: IP Addressing & Subnetting Plan

The network utilizes the 192.168.15.0 Class C block.²²²² To optimize address space, the LANs were assigned /29 masks (8 addresses each) and WAN links were assigned /30 masks (4 addresses each).³³³³

+3

Subnet Allocation Table

Network Segment	Network ID	Subnet Mask	Usable Range	Purpose
LAN A	192.168.15.0	255.255.255.248	.1 – .6	End-user segment A ⁴
LAN B	192.168.15.8	255.255.255.248	.9 – .14	End-user segment B ⁵
LAN C	192.168.15.16	255.255.255.248	.17 – .22	End-user segment C ⁶
LAN D	192.168.15.24	255.255.255.248	.25 – .30	Server & Admin segment ⁷⁷⁷⁷ +1
WAN E	192.168.15.32	255.255.255.252	.33 – .34	Router0 to ISP Link ⁸

WAN F	192.168.15.36	255.255.255.25 2	.37 – .38	Router0 to Router1 Link ⁹
WAN G	192.168.15.40	255.255.255.25 2	.41 – .42	Router1 to Router2 Link ¹⁰
WAN H	192.168.15.44	255.255.255.25 2	.45 – .46	Router2 to Router3 Link 11111111 +1

Section 2: Routing Strategy

The network employs a three-tier routing approach to ensure efficient data flow and internet accessibility:

1. **Static Routing:** Manually defined paths between all internal LAN segments to ensure internal reachability.
2. **Default Routing:** Configured on all edge routers to point toward the ISP (Router "WE") for external traffic handling.
3. **Static Host Routing:** A specific host route was implemented on Router0 to optimize the path to the central Server-PT located in LAN D.

Section 3: Network Services & Application Layer

The infrastructure provides the following centralized services to all end devices:

- **Dynamic Host Configuration Protocol (DHCP):** Each router acts as a DHCP server for its respective LAN, automating IP, Gateway, and DNS assignment.
 - **Domain Name System (DNS):** Hosted on the Server-PT (192.168.15.26) to resolve the company domain and web services.
 - **Web & File Services:**
 - **HTTP:** Internal company web portal.
 - **FTP:** Secure file repository with user-based authentication (e.g., users: ali, body).
 - **Email System:** Configured with a dedicated domain (gmail.com or custom) to support internal corporate communication.
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Section 4: Verification and Results

- **Connectivity:** Full ICMP (Ping) reachability confirmed between all end-devices across the topology.
- **ISP Connectivity:** Internal PCs successfully reached the "WE" ISP router via the default gateway.
- **Service Access:** Verified successful FTP logins, web page rendering, and DNS resolution from multiple LANs.

Would you like me to generate the specific Cisco CLI configuration commands for the DHCP pools or the Static Routes to include in an appendix?